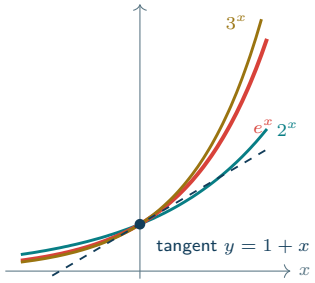
FUNCTIONS LADDER 4 OF 10

Exponential functions and the number e

Quantities that grow in proportion to themselves follow e^x, the exponential whose slope equals its height.

1 / THE RULE AND THE PICTURE

Among all exponentials a^x the one with slope 1 at $x = 0$ is $a = e = 2.71828\ldots$



All the curves pass through $(0, 1)$, and the slope there is $\ln a$: $\ln 2 = 0.693$ for 2^x and $\ln 3 = 1.099$ for 3^x, with e the base that makes the slope exactly 1.
Since the slope of a^x is proportional to its height, e^x is the one whose slope equals its height everywhere:

$$\frac{d}{dx}e^x = e^x, \qquad e^0 = 1, \qquad e^{a+b} = e^a e^b$$

2 / TWO WAYS TO COMPUTE e

A limit of compound interest and a sum of reciprocal factorials

n	$(1 + 1/n)^n$
1	2.00000
2	2.25000
5	2.48832
10	2.59374
100	2.70481
1 000	2.71692
1 000 000	2.71828

Splitting a year into n parts at interest $100\%/n$ gives $(1 + \frac{1}{n})^n$.

terms up to $1/m!$	sum
0	1.00000
1	2.00000
2	2.50000
3	2.66667
4	2.70833
5	2.71667
6	2.71806
7	2.71825

$e = \sum_{k \geq 0} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \cdots$, which converges much faster.

3 / GROWTH AND DECAY

A constant percentage rate gives e^{rt}, with a doubling time of $\ln 2/r$

At 7% per year the doubling time is $\ln 2 / \ln 1.07 = 10.24$ years, close to the rule of 70, $70/7 = 10$, and with continuous compounding it is $\ln 2/0.07 = 9.90$ years.
Investing £1000 at 5% for ten years gives £1628.89 yearly, £1647.01 monthly and $£1648.72 = 1000e^{0.5}$ continuously.
Radioactive decay $N = N_0 e^{-\lambda t}$ with half-life 5730 years for carbon-14 has $\lambda = \ln 2/5730 = 1.21 \times 10^{-4}$ per year.

4 / ANY BASE / $a^x = e^{x \ln a}$

Every exponential is e^x with the axis rescaled

Writing $a = e^{\ln a}$ gives $a^x = e^{x \ln a}$, so $2^x = e^{0.693x}$ and $\frac{d}{dx}a^x = (\ln a)a^x$.
The laws $a^x a^y = a^{x+y}$ and $(a^x)^y = a^{xy}$ follow from $e^{a+b} = e^a e^b$, and the exponential eventually beats every power: at $x = 20$, $e^{20} = 4.85 \times 10^8$ while $20^5 = 3.2 \times 10^6$, a ratio of 0.0066, and the ratio $x^n/e^x \to 0$ for every n.

5 / WHY e / THE DIFFERENTIAL EQUATION

The equation $y' = ky$ has the solution $y = y_0 e^{kt}$

A quantity whose rate of change is proportional to its size, such as a population with unlimited food or a cooling temperature difference, satisfies $y' = ky$, and the exponential is its only solution with $y(0) = y_0$.
Trying $y = e^{rt}$ gives $r = k$, the simplest case of the characteristic equation used for second-order equations in a later sheet.

ASIDE / ARITHMETIC SHORTCUT

For small x, $e^x \approx 1 + x$, so $e^{0.05} \approx 1.05$ against the exact 1.0513.

LINKS TO OTHER SHEETS

The inverse of e^x is in *Logarithms*; the series is developed in *Taylor series*; the previous sheet is *Small-angle approximations*.

Convention. $e = 2.718281828\ldots$; $\ln$ is the natural logarithm; r is a growth rate per unit time; λ a decay rate.